

PHZ / PNZ / SEZ / SEZT

Tubular Casing Pump



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|--|------------------------------|---------------------|
| • Seawater desalination | • Water extraction | • Water supply |
| • Cooling water pumps in power stations and industry | • Dewatering | • Dry docks |
| • Flue gas desulphurisation | • General irrigation systems | • Locks and sluices |
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PHZ / PNZ / SEZ / SEZT – Tubular Casing Pump

Benefits

■ High-quality materials ensure high level of availability

- Each pump is fitted with product-lubricated polymer plastic bearings (Thordon, Greene Tweed).
- Alternatively, a version with maintenance-free and wear-resistant RESIDUR® bearings developed by KSB is available.

■ Efficient water supply thanks to individualised pump selection

- Tailor-made pumps are selected from a modular design system covering a wide range of hydraulic systems, materials and installation types.
- Available with fish-friendly hydraulic systems
- High efficiencies and an excellent NPSH behaviour help reduce the system's life cycle costs.

■ Economically efficient adjustment

- Adjustments to changed system conditions (flow rate/head) can be made by pre-swirl control equipment or variable speed drives (SEZ).
- The axial propeller can be adjusted during operation and standstill (PNZ/PHZ).

■ Optimised pump design reduces maintenance costs

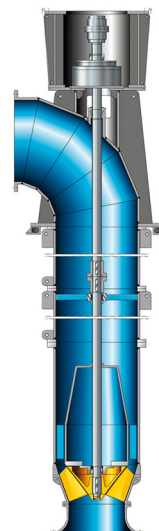
- KSB tubular casing pumps are available in two design variants.
- Pumps in pull-out design enable straightforward removal from the casing and make for low maintenance of all parts subject to wear.

■ Individualised pump design

- The pump can be adapted to existing structures and systems.
- Various special solutions are available, e.g. special elbow angles, sealing elements, pumps arranged horizontally at 45 degrees, and floor mounting.

■ Everything from a single source – with KSB's engineering package

- The pump is characterised by high-quality pump technology.
- An in-house test facility is available for performance tests.
- Support and after-sales service are included in the scope of services.
- Model testing of intake structures is offered.
- Expert consultancy for system planning and project management can be provided.
- Support with planning fish-friendly pump systems



Operating data

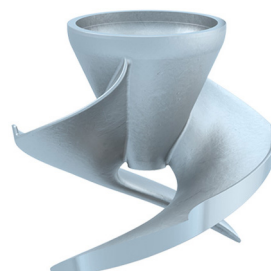
Characteristic		Value			
		SEZ	SEZT	PHZ	PNZ
Flow rate ¹⁾	[m³/h]	≤ 65000	≤ 20000	≤ 65000	≤ 65000
Head ¹⁾	H [m]	≤ 60	≤ 110	≤ 25	≤ 15
Max. fluid temperature ¹⁾	T [°C]	≤ +40	≤ +45	≤ +80	≤ +80
Speed	n [rpm]	≤ 990	≤ 990	≤ 990	≤ 990

Materials

Fluid handled	Component	Material
Fresh water ≤ 28 °C	Casing	Grey cast iron/steel
	Impeller	CrNi steel
	Shaft	Carbon steel
Seawater ≤ 35 °C	Casing	Duplex steel
	Impeller	Duplex steel
	Shaft	Duplex steel
Seawater 35 °C - 40 °C	Casing	Super duplex steel
	Impeller	Super duplex steel
	Shaft	Super duplex steel

In addition, active cathodic protection is available for stainless steel pumps, and sacrificial anodes are available for coated steel pumps.

Fish-friendly pump impeller



¹ Higher heads, flow rates and temperatures on request